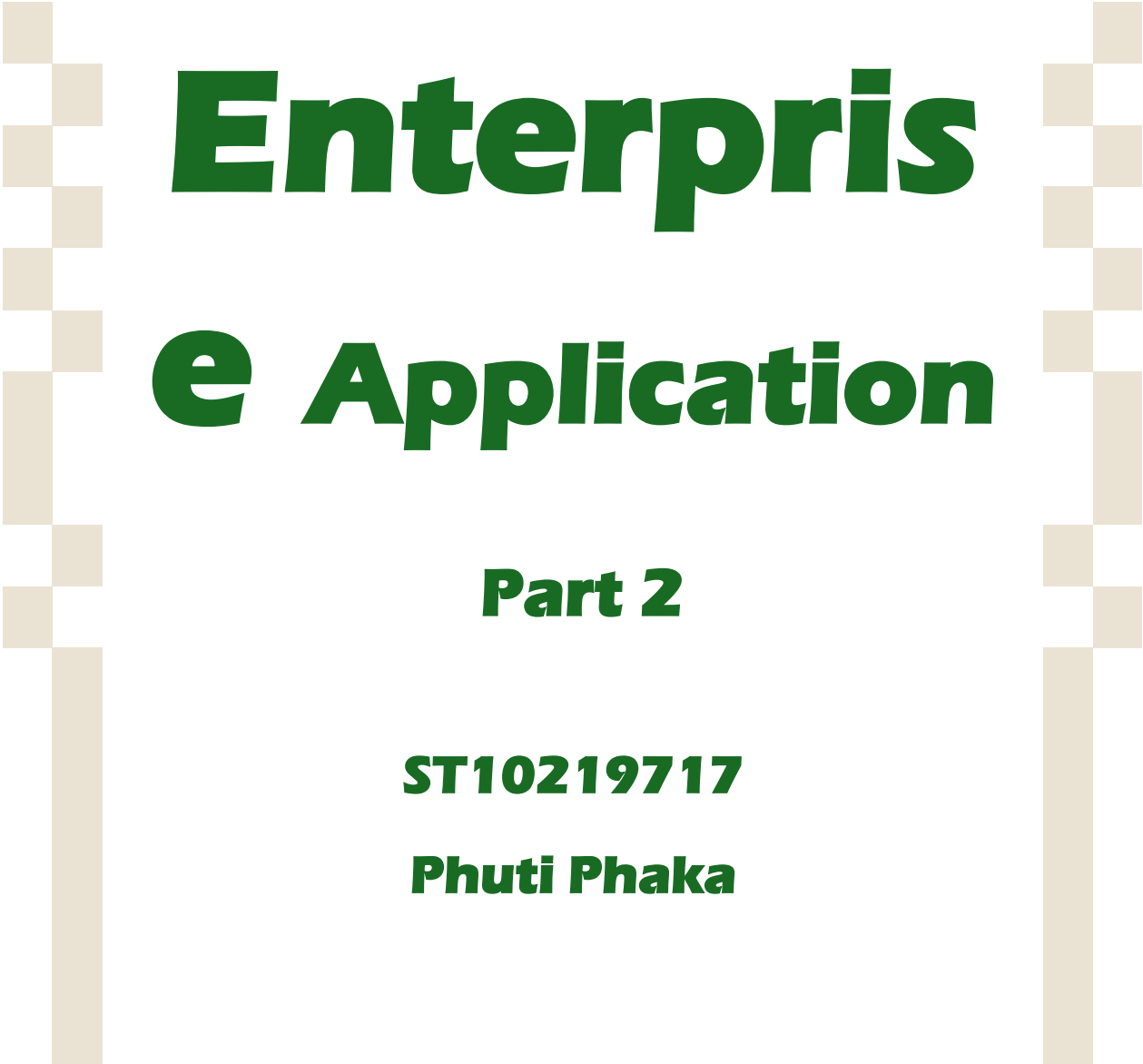




Enterpris e Application

Part 2

ST10219717

Phuti Phaka

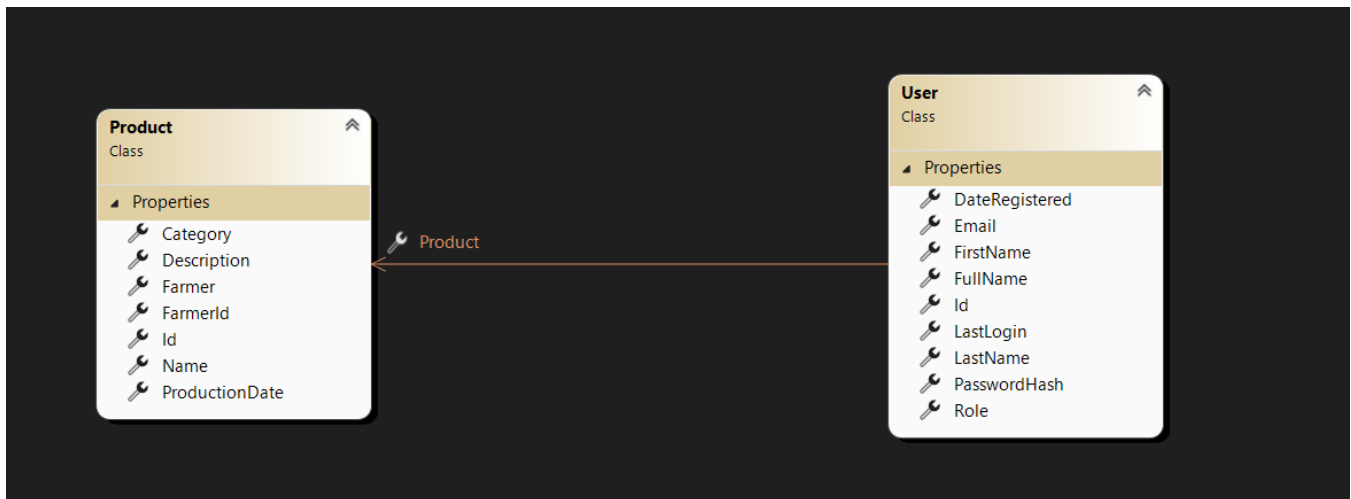
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Agri Connect Platform

DATABASE DEVELOPMENT AND INTEGRATION

Africa Energy Connect is an innovative digital platform designed to transform energy access and management across Africa by bridging the gap between farmers, businesses, and energy providers. With a focus on sustainability and efficiency, the app empowers users to monitor, optimize, and manage energy consumption while promoting renewable energy adoption.

ER Diagram



Populated database with simple data

For Users

Home > academicdata (solver-dev/academicdata)

academicdata (solver-dev/academicdata) | Query editor (preview) ☆ ...

SQL database

Search Login + New Query Open query Feedback Getting started

Overview
Activity log
Tags
Diagnose and solve problems
Query editor (preview)
Mirror database in Fabric (preview)
Resource visualizer
Settings
Data management
Integrations
Power Platform
Security
Intelligent performance
Monitoring
Automation
Help

Query 1 ✕

Run Cancel query Save query Export data as Show only Editor

```
1 select * from [dbo].[Users]
2
3
```

Results Messages

Search to filter items...

Id	FirstName	LastName	Email	PasswordHash	Role	DateRegistered	LastL
1	Thabiso	Phaka	farmer@email.com	\$2a5125x6r7py87kZF11nHDUy8R...	Farmer	2025-04-19T14:04:48.8378363	
2	yoven	smith	yours@email.com	\$2a5125.8/7co.4AyH8tP6M5rzUk...	Farmer	2025-04-19T16:05:40.3640266	
3	Yolanda	Markhams	yolanda@gmail.com	\$2a5125mCxtRfEy.7.25Zbd7LOuZ...	Farmer	2025-05-08T14:43:03.7969094	2025
4	Donald	Featherson	owner@gmail.com	\$2a51257smczjYcmih83CbF0Tcu...	Employee	2025-05-10T18:10:32.1583354	2025

Add or remove favorites by pressing Ctrl+Shift+F

For Products

Home > academicdata (solver-dev/academicdata)

academicdata (solver-dev/academicdata) | Query editor (preview) ☆ ...

SQL database

Search Login + New Query Open query Feedback Getting started

Overview
Activity log
Tags
Diagnose and solve problems
Query editor (preview)
Mirror database in Fabric (preview)
Resource visualizer
Settings
Data management
Integrations
Power Platform
Security
Intelligent performance
Monitoring
Automation
Help

Query 1 ✕

Run Cancel query Save query Export data as Show only Editor

```
1 select * from [dbo].[Products]
2
```

Results Messages

Search to filter items...

Id	Name	Category	ProductionDate	Description	FarmerId
1	banana	Fruits	2025-04-19T00:00:00.0000000	buy out fresh fruits	1
2	Apple	Fruits	2025-04-19T00:00:00.0000000	fresh and affordable	1
3	carrot	Vegetables	2025-04-19T00:00:00.0000000	stay health with our vegetable	1
4	rice	Dairy	2025-04-24T00:00:00.0000000	retfidersrftf	1

Add or remove favorites by pressing Ctrl+Shift+F

System Design

Login

Authentication

The system supports secure login for both **Farmers** and **Employees** using unique credentials.

Example Credentials (for testing purposes only):

- **Farmer**
 - Email: yolanda@gmail.com
 - Password: Yolanda@12
- **Employee**
 - Email: owner@gmail.com
 - Password: Owner@12

⚠ These credentials are for demonstration/testing only. In production, always secure user data using hashing and authentication best practices.

AgricConnect

AgricConnect

Connecting Agriculture & Sustainable Energy

Create an account

Join AgricConnect to start managing your agricultural energy solutions

First Name

Last Name

Email Address

your@email.com

Select Your Role

Farmer

Password

Confirm Password

☐ I agree to the Terms of Service and Privacy Policy

Create Account

Already have an account? [Sign in](#)

The Farmers

Add New Agricultural Products

Farmers can enter product name, type (e.g., fruits, dairy), production date, and a brief description.

AgriConnect - Farmer Dashboard (Yolanda Markhams)

AgriConnect

Welcome, Yolanda Markhams!

Dashboard

Add Product

Search Products

My Profile

Log Out

Add New Product

Product Name

Category

Vegetables

Production Date

2025/05/12

Description

Save Product

View and Manage Products

Farmers can edit or delete their previously added products.

AgriConnect - Farmer Dashboard (Yolanda Markhams)

AgriConnect

Welcome, Yolanda Markhams!

Dashboard

Add Product

Search Products

My Profile

Log Out

My Products

African Mango Fruits Production Date: 2025/05/11 A tropical south african summar fruits	Edit Delete
Orange Fruits Production Date: 2025/05/12 Orange production provides both nutritional and financial benefits to farmers across Africa.	Edit Delete

AgriConnect - Farmer Dashboard (Yolanda Markhams)

AgriConnect

Welcome, Yolanda Markhams!

Dashboard

Add Product

Search Products

My Profile

Log Out

Edit Product

Product Name*
African Mango

Category*
Fruits

Production Date*
2025/05/11

Description
A tropical south african summar fruits

CancelSave

Search Products

Easily locate specific items using the built-in product search feature.

AgriConnect - Farmer Dashboard (Yolanda Markhams)

AgriConnect

Welcome, Yolanda Markhams!

Dashboard

Add Product

Search Products

My Profile

Log Out

african mangoSearch

African Mango

Fruits
Production Date: 2025/05/11
A tropical south african summar fruits

Farmer Details

Farmer can change their personal information including the passwords

AgriConnect - Farmer Dashboard (Yolanda Markhams)

AgriConnect

Welcome, Yolanda Markhams!

My Profile

ProfileFarm Details

Personal Information

First Name:

Yolanda

Last Name:

Markhams

Email:

yolanda@gmail.com

Member Since:

08 May 2025

Account Security

Change Password

Save Changes

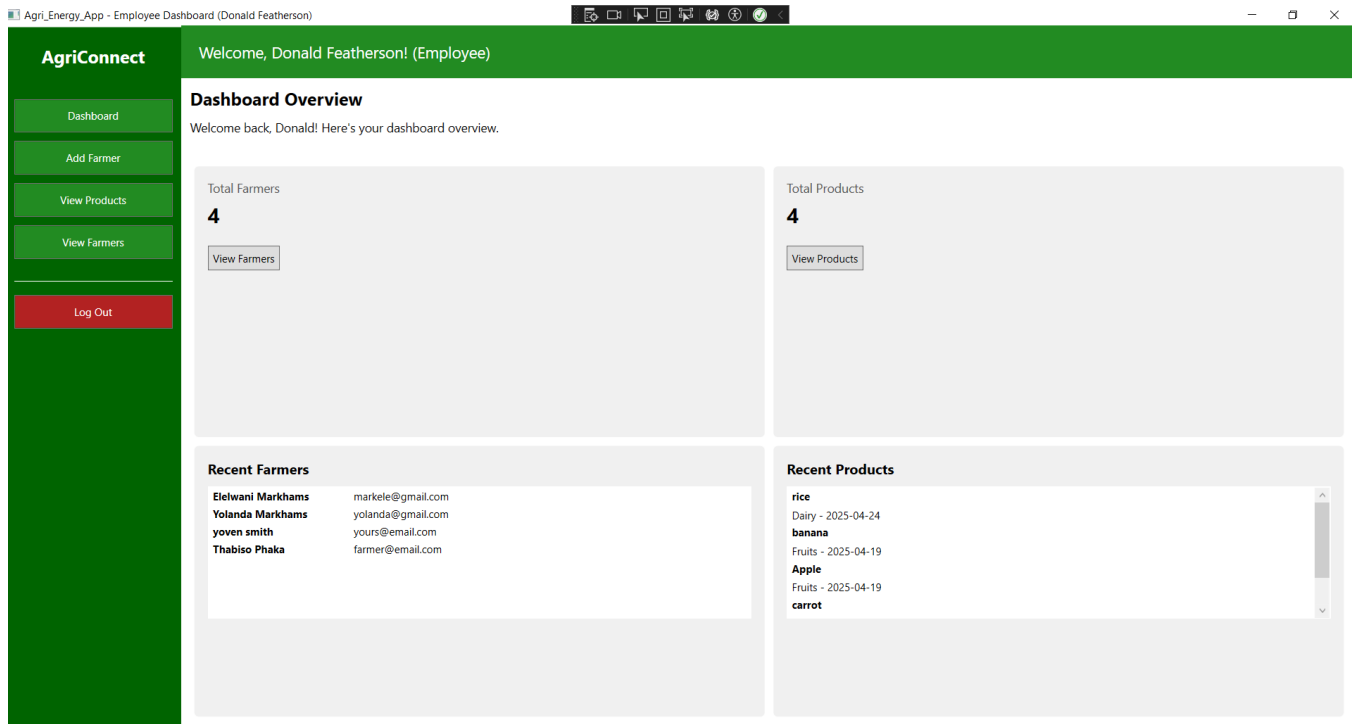
Welcome, Yolanda

Back to Dashboard

Employee

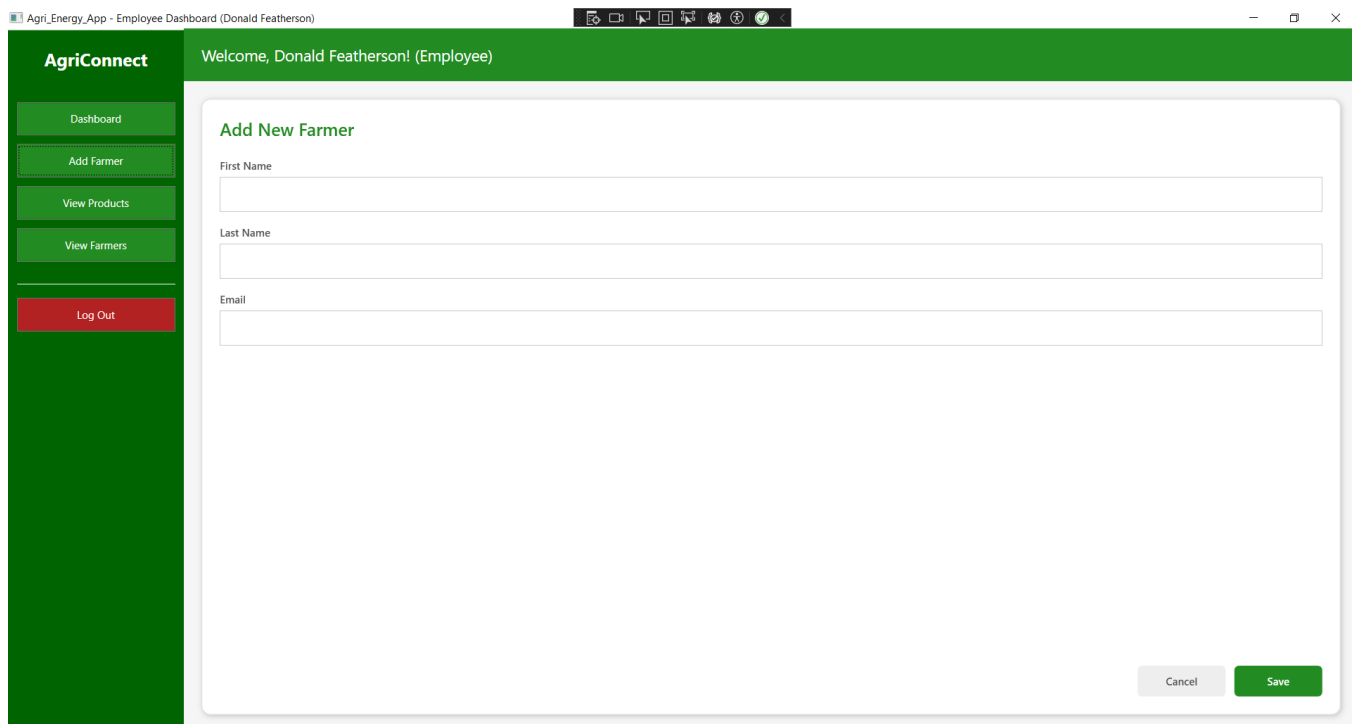
Employee

Employee can access all the farmer with their products



Add New Farmers

Employees can register new farmers into the system



View All Farmers

Easily to see the list of registered farmers along with all their details.

Agri_Energy_App - Employee Dashboard (Donald Featherson)

AgriConnect

Dashboard

Add Farmer

View Products

View Farmers

Log Out

Welcome, Donald Featherson! (Employee)

All Farmers

First Name	Last Name	Email	Date Registered	Last Login
Elelwani	Markhams	markele@gmail.com	2025-05-11	
Yolanda	Markhams	yolanda@gmail.com	2025-05-08	2025-05-12 12:28
Thabiso	Phaka	farmer@email.com	2025-04-19	
yoven	smith	yours@email.com	2025-04-19	

View All Products

Employees can browse all products submitted by any farmer and the product type, also the date

Agri_Energy_App - Employee Dashboard (Donald Featherson)

AgriConnect

Dashboard

Add Farmer

View Products

View Farmers

Log Out

Welcome, Donald Featherson! (Employee)

All Products

Thabiso Phaka

All Categories

Select a date15toSelect a date15

Clear Filters

Product	Category	Production Date	Farmer	Description
Apple	Fruits	2025-04-19		fresh and affordable
banana	Fruits	2025-04-19	Thabiso Phaka	buy out fresh fruits
carrot	Vegetables	2025-04-19	Thabiso Phaka	stay health with our vegetable
rice	Dairy	2025-04-24	Thabiso Phaka	retfdedsrftf

ERROR HANDLING



Welcome back

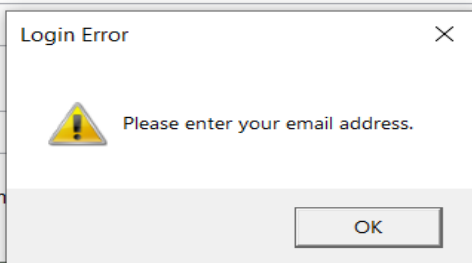
Enter your credentials to access your account

Email Address

Login Error

Password

 Please enter your email address.

☐ Remember m

Sign in

Don't have an account? [Sign up](#)

Empowering farmers with sustainable energy solutions for a greener tomorrow.
— AgricConnect Vision

AgricConnect

Connecting Agriculture & Sustainable Energy

Welcome back

Enter your credentials to access your account

Email Address

KNNKDNK

Password

☐ Remember me

Login Error



Please enter a valid email address.

OK

Sign in

Don't have an account? [Sign up](#)

Empowering farmers with sustainable energy solutions for a greener tomorrow.

— AgricConnect Vision

AgricConnect

Connecting Agriculture & Sustainable Energy

Welcome back

Enter your credentials to access your account

Email Address

Password

☐ Remember me for next time

Sign in

Don't have an account? [Sign up](#)

Login Error



Invalid email or password.

OK

Create an account

Join AgricConnect to start managing your agricultural energy solutions

First Name

Last Name

Email Address

Select Your Role

Farmer

▼

Password

Confirm Password

☐ I agree to the Terms of Service and Privacy Policy

Create Account

Registration Error



Please enter your first and last name.

OK

Join AgriConnect to start managing your agricultural energy solutions

First Name

malebo

Last Name

smith

Email Address

your@email.com

Select

Farm

Password

•••••

Confirm Password

••••••

Registration Error



Password must contain at least one letter and one number.

OK

Email Address

our@email.com

Select Your Role

irmer

Password

••••••••

Confirm Password

•••••

Registration Error



Passwords do not match.

OK

I agree to the Terms of Service and Privacy Policy

malebo

smith

Email Address

your@email.com

Select

Farm

Password

••••

Confirm Password

••••••••

☐ I agree to the Terms of Service and Privacy Policy

Create Account

Registration Error



You must agree to the Terms of Service and Privacy Policy.

OK

AgricConnect

Connecting Agriculture & Sustainable Energy

Create an account

Join AgricConnect to start managing your agricultural energy solutions

First Name

malebo

Last Name

smith

Email Address

your@email.com

Select Your Role

Farmer

Password

••••

Confirm Password

••••••

☐ I agree to the Terms of Service and Privacy Policy

Registration Error



Password must be at least 8 characters long.

OK

Add New Product

Product Name

Category

Vegetables

Production Date

2025/05/12

Description

Save Product

Error

 Please enter a product name

OK

Link for GIT

[phutithabiso/Agri-Energy-App](https://github.com/phutithabiso/Agri-Energy-App)

